# Distributed Ledger Intelligence Framework for End-to-End Agriculture Food Supply Traceability Transparency Scure Governance

Project Phase-I Report submitted to Osmania University, Hyderabad, in partial fulfillment of the requirement for the award of the degree of

**BACHELOR OF ENGINEERING**

in

**INFORMATION TECHNOLOGY**

By


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Under the Guidance of


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Assistant Professor






**DEPARTMENT OF INFORMATION TECHNOLOGY**
**LORDS INSTITUTE OF ENGINEERING AND TECHNOLOGY**
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**Himayathsagar, Hyderabad,500091**


**A.Y. 2026-2027**



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## CERTIFICATE

This is to certify that the Project Phase-I report entitled "**Distributed Ledger Intelligence Framework for End-to-End Agriculture Food Supply Traceability Transparency Scure Governance"** being submitted by **Mr. Azam Amaan Khan (160923737079), Mr. Amer Anus Thander (160923737080), Mr. Mohammed Fazeeluddin (160923737085)** in partial fulfillment of the requirements for the award of the degree of **Bachelor of Engineering** in **Information Technology** is a record of bonafide work carried out by them.

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## DECLARATION BY THE CANDIDATES

**We,** Mr. Azam Amaan Khan (160923737079), Mr. Amer Anus Thander(160923737080) and Mr. Mohammed Fazeeluddin (160923737085), hereby declare that the Project Phase-I report entitled "**Distributed Ledger Intelligence Framework for End-to-End Agriculture Food Supply Traceability Transparency Scure Governance"** is carried out under the guidance of **Ms. P Srilatha**, Assistant Professor, Department of Information Technology, **Lords Institute of Engineering and Technology**, **Hyderabad** is submitted in partial fulfillment of the requirements for the award of the degree of **Bachelor of Engineering** in **Information Technology.**

This is a record of bonafide work carried out by us and the results embodied in this project have not been reproduced or copied from any source. The results embodied in this project report have not been submitted to any other university or institute for the award of any other degree or diploma.

**Azam Amaan Khan (160923737079)**
**Amer Anus Thander (160923737080)**
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# ACKNOWLEDGEMENT

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**Azam Amaan Khan (160923737079)**
**Amer Anus Thander (160923737080)**
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**Vision of the Department:**

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**DM2:** Provide learner centric environment to enrich knowledge, skills and innovation through industry academia collaborations.

**DM3:** Train the concerned stakeholders on latest technologies to meet the industry needs.

**DM4:** Inculcate leadership, professional, interpersonal skills and ethical values to address the contemporary societal issues.

**Note:** DM: Department Mission

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### B.E Information Technology Program Educational Objectives (PEOs):

| | |
|---|---|
| **PEO1** | Be a competent software engineer in IT and allied industry providing viable solutions. |
| **PEO2** | Exhibit professionalism, engage in lifelong learning and adopt to changing industry and societal needs. |
| **PEO3** | Be effective as an individual and also cohesive in a group to execute multidisciplinary projects. |
| **PEO4** | Pursue higher studies in Information Technology with emphasis on competency and innovation. |

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### B.E Information Technology Program Outcomes (POs): Engineering

### Graduate will be able to:

| S.no | Program Outcomes (POs) |
|---|---|
| 1. | **PO1: Engineering Knowledge:** Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop to the solution of complex engineering problems. |
| 2. | **PO2: Problem Analysis:** Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development. (WK1 to WK4) |
| 3. | **PO3: Design/Development of Solutions:** Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required. (WK5) |
| 4. | **PO4: Conduct Investigations of Complex Problems:** Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions. (WK8). |
| 5. | **PO5: Engineering Tool Usage:** Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems. (WK2 and WK6) |
| 6. | **PO6: The Engineer and The World:** Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment. (WK1, WK5, and WK7). |
| 7. | **PO7: Ethics:** Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws. (WK9) |
| 8. | **PO8: Individual and Collaborative Team work:** Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams. |
| 9. | **PO9: Communication:** Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences |
| 10. | **PO10: Project Management and Finance:** Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments. |
| 11. | **PO11: Life-Long Learning:** Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change. (WK8) |



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### B.E Information Technology Program Specific Outcomes (PSOs):

| | |
|---|---|
| **PSO1** | **Professional Skills:** Identify, analyze requirements, design and implement computer programs in the area of artificial intelligence, machine learning, big data analytics and networking of varying complexity. |
| **PSO2** | **Problem-Solving Skills:** Deliver quality service to IT industry and provide solutions to real-life problems applying modern computer languages, software methods and tools. |

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**Course Outcomes: Project Phase-I**
**After completion of the project, the student will be able to**

| S.No | Description | Blooms Taxonomy Level |
|---|---|---|
| PP.I.CO1 | Demonstrate the ability to synthesize and apply the knowledge and skills acquired in the academic program to the real-world problems. | BTL3 |
| PP.I.CO2 | Evaluate different solutions based on economic and technical feasibility | BTL4 |
| PP.I.CO3 | Effectively plan a project and confidently perform all aspects of project management. | BTL5 |
| PP.I.CO4 | Demonstrate effective written and oral communication skills | BTL4 |
| PP.I.CO5 | Prepare the documentation report and perform the presentation of the project work | BTL3 |

**Note: Bloom's Taxonomy Levels**

| | | |
|---|---|---|
| BTL1-Remember | BTL2 – Understand | BTL3 –Apply |
| BTL4-Analyze | BTL5 –Evaluate | BTL6–Create |

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**Course Articulation Matrix:**

**Mapping of Course Outcomes (COs) with Program Outcomes (POs) and Program Specific Outcomes (PSOs):**

| Course Outcomes (COs) | Program Outcomes (POs) | | | | | | | | | | | Program Specific Outcomes (PSOs) | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO 10 | PO 11 | PSO1 | PSO2 |
| PP.I.CO1 | 3 | 2 | 2 | - | 2 | 2 | - | - | 3 | - | - | 2 | 2 |
| PP.I.CO2 | 3 | 3 | 2 | 3 | 2 | - | 2 | - | 3 | - | - | - | 2 |
| PP.I.CO3 | 3 | 3 | 3 | - | 3 | 3 | 1 | 2 | 2 | 3 | 2 | 3 | - |
| PP.I.CO4 | 3 | 2 | 2 | 3 | 3 | 3 | 3 | 2 | - | - | 2 | 3 | 3 |
| PP.I.CO5 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 2 | 3 | 3 | - | 3 | 3 |
| AVG | **3** | **2.6** | **2.5** | **3** | **2.6** | **2.7** | **2.2** | **2** | **2.9** | **3** | **2** | **2.7** | **2.5** |

**Level:**

1- Low correlation (Low), 2- Medium correlation (Medium), 3-High correlation (High)

**SDG Mapping:**

| SDG | Mapped Indicator | SDG | Mapped Indicator | SDG | Mapped Indicator |
|---|---|---|---|---|---|
| 1 NO POVERTY | | 7 AFFORDABLE AND CLEAN ENERGY | | 13 CLIMATE ACTION | |
| 2 ZERO HUNGER | ✓ | 8 DECENT WORK AND ECONOMIC GROWTH | | 14 LIFE BELOW WATER | |
| 3 GOOD HEALTH AND WELL-BEING | | 9 INDUSTRY, INNOVATION AND INFRASTRUCTURE | ✓ | 15 LIFE ON LAND | |
| 4 QUALITY EDUCATION | | 10 REDUCED INEQUALITIES | | 16 PEACE, JUSTICE AND STRONG INSTITUTIONS | ✓ |
| 5 GENDER EQUALITY | | 11 SUSTAINABLE CITIES AND COMMUNITIES | | 17 PARTNERSHIPS FOR THE GOALS | |
| 6 CLEAN WATER AND SANITATION | | 12 RESPONSIBLE CONSUMPTION AND PRODUCTION | ✓ | | |

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# ABSTRACT

The traditional agricultural food supply chain faces several challenges, including lack of transparency, difficulty in tracking products, limited information sharing, and the risk of fraudulent or altered records. These issues make it difficult for consumers to verify the origin and journey of agricultural products and reduce accountability among supply chain participants. To address these challenges, the proposed AgriChain system uses blockchain technology to improve transparency, traceability, and trust in the agricultural food supply chain.

AgriChain uses blockchain technology with Polygon/Ethereum compatibility and smart contracts to maintain transparent and tamper-resistant records of product movements and transactions. The system provides role-based access for farmers, distributors, retailers, consumers, and administrators. Farmers can create product batches, distributors can update shipment details, and retailers can confirm product receipt and update product availability. Consumers can scan QR codes to access product information and trace its journey from the farm to the retail store. The administrator manages user access and helps ensure that only authorized users can perform designated operations.

By maintaining a shared record of supply chain activities, AgriChain aims to reduce information gaps, improve accountability, and make product traceability more accessible. The system provides a practical approach to improving agricultural supply chain management through blockchain-based record keeping and QR-code verification. It can also serve as a foundation for future enhancements, such as IoT-based product condition monitoring, automated payments, and advanced analytics.

**Keywords:** Blockchain, AgriChain, Polygon, Smart Contracts, Agricultural Food Supply Chain, Traceability, Transparency, QR Code.

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